

Embedded TCP/IP Network Device Manager

Program Output Screenshots

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1. Source code in nano: device.cpp (left) and server.cpp (right)

~/.EmbeddedNetworkMonitor/device

GNU nano 9.2device.cpp

```
#include <iostream>
#include <cstring>
#include <winsock2.h>

#pragma comment(lib, "ws2_32.lib")

int main()
{
    std::cout << "Starting embedded device..." << std::endl;

    // 1. Initialize Winsock
    WSADATA wsaData;

    int result = WSASStartup(MAKEWORD(2, 2), &wsaData);

    if (result != 0)
    {
        std::cout << "WSAStartup failed." << std::endl;
        return 1;
    }

    std::cout << "Winsock initialized successfully." << std::endl;

    // 2. Create TCP socket
    SOCKET clientSocket = socket(
        AF_INET,
        SOCK_STREAM,
        IPPROTO_TCP
    );

    if (clientSocket == INVALID_SOCKET)
    {
        std::cout << "Socket creation failed." << std::endl;
        WSACleanup();
        return 1;
    }

    std::cout << "TCP socket created successfully." << std::endl;

    // 3. Configure server address
    sockaddr_in serverAddress;

    serverAddress.sin_family = AF_INET;
    serverAddress.sin_addr.s_addr = inet_addr("127.0.0.1");
    serverAddress.sin_port = htons(5000);

    // 4. Connect to server
    result = connect(
        clientSocket,
        reinterpret_cast<sockaddr*>(&serverAddress),
        sizeof(serverAddress)
    );

    if (result == SOCKET_ERROR)
    {
        std::cout << "Connection to server failed." << std::endl;
        closesocket(clientSocket);
    }
}
```

[Read 77 lines]

AG Help AO Write Out AF Where Is AK Cut AT Execute AC Location M-U Undo
AX Exit AR Read File AL Replace AU Paste AJ Justify AN Go To Line M-E Redo

~/.EmbeddedNetworkMonitor/server

GNU nano 9.2server.cpp

```
#include <iostream>
#include <winsock2.h>

#pragma comment(lib, "ws2_32.lib")

int main()
{
    std::cout << "Starting TCP Server..." << std::endl;
    WSADATA wsadata;

    int result=WSASStartup(MAKEWORD(2,2),&wsadata);

    if(result!=0)
    {
        std::cout << "WSAStartup failed." << std::endl;
        return 1;
    }

    std::cout << "Winsock initalized syccessfully." << std::endl;
    SOCKET serverSocket = socket(AF_INET,SOCK_STREAM,IPPROTO_TCP);

    if(serverSocket == INVALID_SOCKET)
    {
        std::cout << "Socket Creation Faile." << std::endl;
        WSACleanup();
        return 1;
    }

    std::cout << "TCP socket created successfully." <<std::endl;
    sockaddr_in serverAddress;

    serverAddress.sin_family=AF_INET;
    serverAddress.sin_addr.s_addr=inet_addr("127.0.0.1");
    serverAddress.sin_port=htons(5000);

    result =bind(serverSocket,reinterpret_cast<sockaddr*>(&serverAddress),sizeof(serverAddress));

    if (result == SOCKET_ERROR)
    {
        std::cout << "Bind Failed." << std::endl;
        closesocket(serverSocket);
        WSACleanup();
        return 1;
    }

    std::cout << "Server bound to 127.0.0.1:5000" << std::endl;

    result=listen(serverSocket,5);
    if(result == SOCKET_ERROR)
    {
        std::cout << "listen Failed..." << std::endl;
        closesocket(serverSocket);
        WSACleanup();
        return 1;
    }

    SOCKET clientSocket=accept(serverSocket,nullptr,nullptr);
}
```

[Read 72 lines]

AG Help AO Write Out AF Where Is AK Cut AT Execute AC Location M-U Undo
AX Exit AR Read File AL Replace AU Paste AJ Justify AN Go To Line M-E Redo

2. Compiling with g++ and starting the server and device

```
~/EmbeddedNetworkMonitor/device
acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ ./device.exe
Starting embedded device...
Winsock initialized successfully.
TCP socket created successfully.
Connected to TCP server!
Data sent to server.

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ nao device.cpp
-bash: nao: command not found

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ nano device.cpp

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ rm device.exe

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ nano device.cpp

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ rm device.cpp

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ nano device.cpp

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ ./device.exe
-bash: ./device.exe: No such file or directory

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ cd ~/EmbeddedDeviceMonitor/device
-bash: cd: /home/acer/EmbeddedDeviceMonitor/device: No such file or directory

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ ls
device.cpp

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ g++ device.cpp -o device.exe -lws2_32

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ ls
device.cpp  device.exe

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ ./device.exe
=====
Embedded Device Simulator
=====
[OK] Winsock initialized.
[OK] TCP socket created.
[CONNECTED] Connected to server.
[SENT] Device registration.
[ERROR] Telemetry transmission failed.
Device shutdown complete.

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ ./device.exe

~/EmbeddedNetworkMonitor/server
$ rm server.cpp

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ nano server.cpp

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ server/server.cpp
-bash: server/server.cpp: No such file or directory

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ ./server.exe
Starting TCP Server...
Winsock initialized successfully.
TCP socket created successfully.
Server bound to 127.0.0.1:5000
Client connected successfully!

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ rm server.cpp

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ nano server.cpp

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ ./server.exe
Starting TCP Server...
Winsock initialized successfully.
TCP socket created successfully.
Server bound to 127.0.0.1:5000
Client connected successfully!

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ g++ server.cpp -o server.exe lws2_32
C:/msys64/ucrt64/bin/./lib/gcc/x86_64-w64-mingw32/16.2.0/./../x86_64-w64-mingw32/bin/ld.exe: ca
nnot find lws2_32: No such file or directory
collect2.exe: error: ld returned 1 exit status

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ ^C

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ ls
server.cpp  server.cpp.save

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ g++ server.cpp -o server.exe -lws2_32

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ ls
server.cpp  server.cpp.save  server.exe

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ ./server.exe
=====
Embedded TCP/IP Network Device Manager
=====
[OK] Winsock initialized.
[OK] TCP socket created.
[OK] Server bound to 127.0.0.1:5000
[OK] Server listening...
```

3. Device registration and telemetry received by the server

```
~/EmbeddedNetworkMonitor/device
-bash: nao: command not found

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ nano device.cpp

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ rm device.exe

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ nano device.cpp

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ rm device.cpp

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ nano device.cpp

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ ./device.exe
-bash: ./device.exe: No such file or directory

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ cd ~/EmbeddedDeviceMonitor/device
-bash: cd: /home/acer/EmbeddedDeviceMonitor/device: No such file or directory

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ ls
device.cpp

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ g++ device.cpp -o device.exe -lws2_32

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ ls
device.cpp  device.exe

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ ./device.exe
=====
Embedded Device Simulator
=====
[OK] Winsock initialized.
[OK] TCP socket created.
[CONNECTED] Connected to server.
[SENT] Device registration.
[ERROR] Telemetry transmission failed.
Device shutdown complete.

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ ./device.exe
=====
Embedded Device Simulator
=====
[OK] Winsock initialized.
[OK] TCP socket created.
[CONNECTED] Connected to server.
[SENT] Device registration.
[SENT] Telemetry -> Temp=28.7 C, Speed=1250 RPM
[SENT] Telemetry -> Temp=28.9 C, Speed=1300 RPM
[SENT] Heartbeat.

~/EmbeddedNetworkMonitor/server
acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ ./server.exe
Starting TCP Server...
Winsock initialized successfully.
TCP socket created successfully.
Server bound to 127.0.0.1:5000
Client connected successfully!

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ g++ server.cpp -o server.exe lws2_32
C:/msys64/ucrt64/bin/./lib/gcc/x86_64-w64-mingw32/16.2.0/./././././x86_64-w64-mingw32/bin/ld.exe: ca
nnot find lws2_32: No such file or directory
collect2.exe: error: ld returned 1 exit status

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ AC

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ ls
server.cpp  server.cpp.save

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ g++ server.cpp -o server.exe -lws2_32

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ ls
server.cpp  server.cpp.save  server.exe

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$ ./server.exe
=====
Embedded TCP/IP Network Device Manager
=====
[OK] Winsock initialized.
[OK] TCP socket created.
[OK] Server bound to 127.0.0.1:5000
[OK] Server listening...
[CONNECTED] Device connected.

[REGISTER]
Device ID   : DEV001
Device Type : ECU
Status      : ONLINE

[TELEMETRY]
Device ID   : DEV001
Temperature : 28.700000 C
Speed       : 1250.000000 RPM
Voltage     : 3.300000 V

[TELEMETRY]
Device ID   : DEV001
Temperature : 28.900000 C
Speed       : 1300.000000 RPM
Voltage     : 3.300000 V

[HEARTBEAT]
Device ID   : DEV001
Status      : ALIVE
```

4. Continuous telemetry and heartbeat; device and server running

```
~/EmbeddedNetworkMonitor/device
acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ ls
device.cpp  device.exe
acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ ./device.exe
=====
Embedded Device Simulator
=====
[OK] Winsock initialized.
[OK] TCP socket created.
[CONNECTED] Connected to server.
[SENT] Device registration.
[ERROR] Telemetry transmission failed.
Device shutdown complete.

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$ ./device.exe
=====
Embedded Device Simulator
=====
[OK] Winsock initialized.
[OK] TCP socket created.
[CONNECTED] Connected to server.
[SENT] Device registration.
[SENT] Telemetry -> Temp=28.7 C, Speed=1250 RPM
[SENT] Telemetry -> Temp=28.9 C, Speed=1300 RPM
[SENT] Heartbeat.
[SENT] Telemetry -> Temp=29.1 C, Speed=1350 RPM
[SENT] Telemetry -> Temp=29.3 C, Speed=1400 RPM
[SENT] Heartbeat.
[SENT] Telemetry -> Temp=29.5 C, Speed=1450 RPM
[SENT] Telemetry -> Temp=29.7 C, Speed=1500 RPM
[SENT] Heartbeat.
[SENT] Telemetry -> Temp=29.9 C, Speed=1550 RPM
[SENT] Telemetry -> Temp=30.1 C, Speed=1600 RPM
[SENT] Heartbeat.
[SENT] Telemetry -> Temp=30.3 C, Speed=1650 RPM
[SENT] Telemetry -> Temp=30.5 C, Speed=1700 RPM
[SENT] Heartbeat.
[SENT] Telemetry -> Temp=30.7 C, Speed=1750 RPM
[SENT] Telemetry -> Temp=30.9 C, Speed=1800 RPM
[SENT] Heartbeat.
[SENT] Telemetry -> Temp=31.1 C, Speed=1850 RPM
[SENT] Telemetry -> Temp=31.3 C, Speed=1900 RPM
[SENT] Heartbeat.
[SENT] Telemetry -> Temp=31.5 C, Speed=1950 RPM
[SENT] Telemetry -> Temp=31.7 C, Speed=2000 RPM
[SENT] Heartbeat.
[SENT] Telemetry -> Temp=31.9 C, Speed=2050 RPM
[SENT] Telemetry -> Temp=32.1 C, Speed=2100 RPM
[SENT] Heartbeat.
[SENT] Telemetry -> Temp=32.3 C, Speed=2150 RPM
[SENT] Telemetry -> Temp=32.5 C, Speed=2200 RPM
[SENT] Heartbeat.
[ERROR] Telemetry transmission failed.
Device shutdown complete.

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/device
$

~/EmbeddedNetworkMonitor/server
acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
[TELEMETRY]
Device ID : DEV001
Temperature : 31.300000 C
Speed : 1900.000000 RPM
Voltage : 3.300000 V

[HEARTBEAT]
Device ID : DEV001
Status : ALIVE

[TELEMETRY]
Device ID : DEV001
Temperature : 31.500000 C
Speed : 1950.000000 RPM
Voltage : 3.300000 V

[TELEMETRY]
Device ID : DEV001
Temperature : 31.700000 C
Speed : 2000.000000 RPM
Voltage : 3.300000 V

[HEARTBEAT]
Device ID : DEV001
Status : ALIVE

[TELEMETRY]
Device ID : DEV001
Temperature : 31.900000 C
Speed : 2050.000000 RPM
Voltage : 3.300000 V

[TELEMETRY]
Device ID : DEV001
Temperature : 32.100000 C
Speed : 2100.000000 RPM
Voltage : 3.300000 V

[HEARTBEAT]
Device ID : DEV001
Status : ALIVE

[TELEMETRY]
Device ID : DEV001
Temperature : 32.300000 C
Speed : 2150.000000 RPM
Voltage : 3.300000 V

[TELEMETRY]
Device ID : DEV001
Temperature : 32.500000 C
Speed : 2200.000000 RPM
Voltage : 3.300000 V

[HEARTBEAT]
Device ID : DEV001
Status : ALIVE

acer@Adharsh UCRT64 ~/EmbeddedNetworkMonitor/server
$
```